Chapter 5

Overcurrent protection

5.1 General

Very high current levels in electrical power systems are usually caused by faults on the system. These currents can be used to determine the presence of faults and operate protection devices, which can vary in design depending on the complexity and accuracy required. Among the more common types of protection are thermo-magnetic switches, moulded-case circuit breakers (MCCBs), fuses and overcurrent relays. The first two types have simple operating arrangements and are principally used in the protection of low-voltage equipment. Fuses are also often used at low voltages, especially for protecting lines and distribution transformers.

Overcurrent relays, which form the basis of this chapter, are the most common form of protection used to deal with excessive currents on power systems. They should not be installed purely as a means of protecting systems against overloads – which are associated with the thermal capacity of machines or lines – since over-current protection is primarily intended to operate only under fault conditions. However, the relay settings that are selected are often a compromise in order to cope with both overload and overcurrent conditions.

5.2 Types of overcurrent relay

On the basis of the relay operating characteristics, overcurrent relays can be classified into three groups: definite current or instantaneous, definite time and inverse time. The characteristic curves of these three types are shown in Figure 5.1, which also illustrates the combination of an instantaneous relay with one having an inverse time characteristic.

5.2.1 Definite-current relays

This type of relay operates instantaneously when the current reaches a predetermined value. The setting is chosen so that, at the substation furthest away from the source, the relay will operate for a low current value and the relay operating currents are progressively increased at each substation, moving towards the source. Thus, the relay with the lower setting operates first and disconnects load at the point nearest to the fault. This type of protection has the drawback of having little selectivity at high values of short-circuit current. Another disadvantage is the

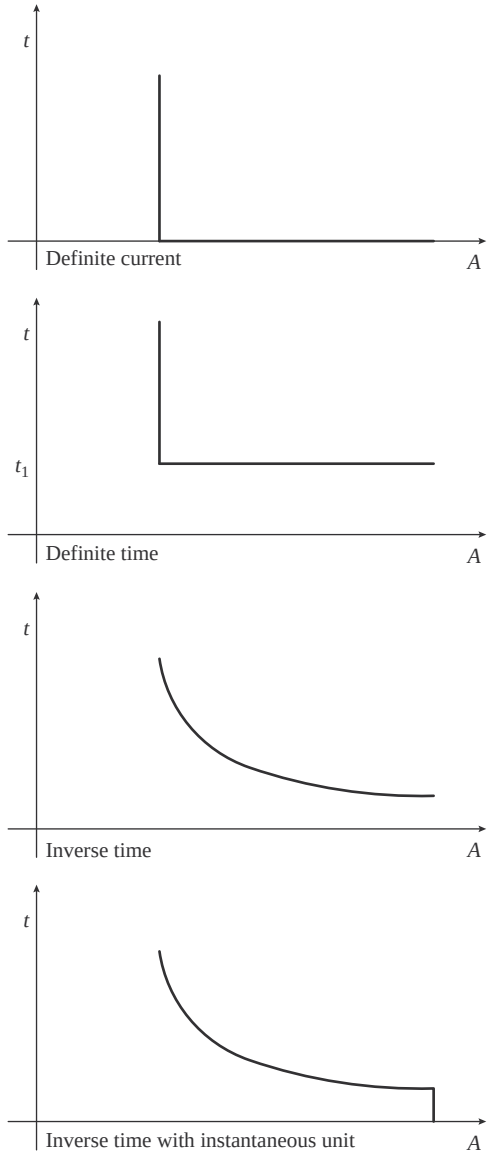


Figure 5.1 Time/current operating characteristics of overcurrent relays

difficulty of distinguishing between the fault current at one point or another when the impedance between these points is small in comparison to the impedance back to the source, leading to the possibility of poor discrimination.

Figure 5.2a illustrates the effect of the source impedance on the short-circuit level at a substation, and for a fault at point B down the line. From Figure 5.2b, it

can be appreciated that the fault currents at F_1 and F_2 are almost the same, and it is this which makes it difficult to obtain correct settings for the relays. When there is some considerable impedance between F_1 and F_2, for example when the fault F_1 is located down a long line, then the fault current at F_1 will be less than that at F_2. Similarly, due to the impedance of the transformer, there will be a considerable difference between the currents for faults at F_2 and F_3, even though these two points are physically close.

If the protection settings are based on maximum fault level conditions, then these settings may not be appropriate for the situation when the fault level is lower. However, if a lower value of fault level is used when calculating the relay settings,

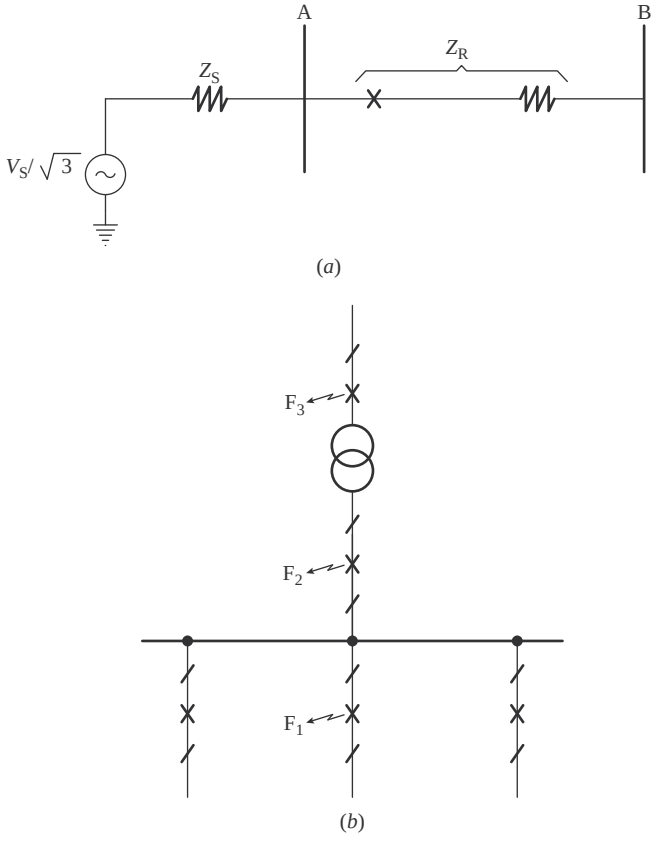


Figure 5.2 (a and b) Illustration of different levels of fault current

Z_R = Impedance of protected element
Z_s = Source impedance

$$I_{SC(A)} = \frac{V_S}{\sqrt{3}} \frac{1}{Z_s}, \quad I_{SC(B)} = \frac{V_S}{\sqrt{3}} \frac{1}{Z_s + Z_R}$$

this could result in some breakers operating unnecessarily if the fault level increases. As a consequence, definite-current relays are not used as the only overcurrent protection, but their use as an instantaneous unit is common where other types of protection are in use.

5.2.2 Definite-time/current or definite-time relays

This type of relay enables the setting to be varied to cope with different levels of current by using different operating times. The settings can be adjusted in such a way that the breaker nearest to the fault is tripped in the shortest time, and then the remaining breakers are tripped in succession using longer time delays, moving back towards the source. The difference between the tripping times for the same current is called the discrimination margin.

Since the operating time for definite-current relays can be adjusted in fixed steps, the protection is more selective. The big disadvantage with this method of discrimination is that faults near to the source, which result in bigger currents, may be cleared in a relatively long time. This type of relay has a current or pick-up (PU) setting – also known as the plug or tap setting – to select the value at which the relay will start, plus a time dial setting to obtain the exact timing of the relay operation. It should be noted that the time-delay setting is independent of the value of the overcurrent required to operate the relay. These relays are used a great deal when the source impedance is large compared to that of the power system element being protected when fault levels at the relay position are similar to those at the end of the protected element.

5.2.3 Inverse-time relays

The fundamental property of these relays is that they operate in a time that is inversely proportional to the fault current, as illustrated by the characteristic curves shown later. Their advantage over definite-time relays is that, for very high currents, much shorter tripping times can be obtained without risk to the protection selectivity. Inverse-time relays are generally classified in accordance with their characteristic curve that indicates the speed of operation; based on this, they are commonly defined as being inverse, very inverse or extremely inverse. Inverse-time relays are also referred to as inverse definite minimum time (IDMT) overcurrent relays.

5.3 Setting overcurrent relays

Overcurrent relays are normally supplied with an instantaneous element and a time-delay element within the same unit. When electromechanical relays were more popular, the overcurrent protection was made up from separate single-phase units. The more modern microprocessor protection has a three-phase overcurrent unit and an earth-fault unit within the same case. Setting overcurrent relays involves selecting the parameters that define the required time/current characteristic of both the time-delay and instantaneous units. This process has to be carried out twice,

once for the phase relays and then repeated for the earth-fault relays. Although the two processes are similar, the three-phase short-circuit current should be used for setting the phase relays while the phase-to-earth fault current should be used for the earth-fault relays. When calculating the fault currents, the power system is assumed to be in its normal operating state. However, at a busbar that has two or more transformers connected in parallel and protected with relays that do not have the facility of multiple setting groups – the ability to be adjusted to accommodate the prevailing system conditions, which is possible with numerical relays, for example – better discrimination is obtained if the calculations are carried out on the basis of each one of the transformers being out of service in turn. The same procedure can be applied to multiple circuit arrangements.

5.3.1 Setting instantaneous units

Instantaneous units are more effective when the impedances of the power system elements being protected are large in comparison to the source impedance, as indicated earlier. They offer two fundamental advantages:

- They reduce the operating time of the relays for severe system faults.
- They avoid the loss of selectivity in a protection system consisting of relays with different characteristics; this is obtained by setting the instantaneous units so that they operate before the relay characteristics cross, as shown in Figure 5.3.

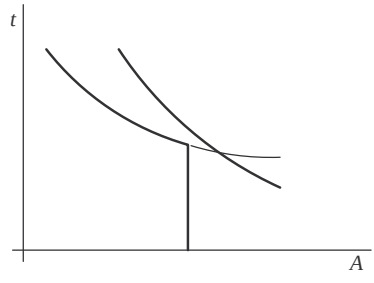


Figure 5.3 Preservation of selectivity using instantaneous units

The criteria for setting instantaneous units vary depending on the location and the type of system element being protected. Three groups of elements can be defined – lines between substations, distribution lines and transformers.

(i) Lines between substations

The setting of instantaneous units is carried out by taking at least 125% of the symmetrical r.m.s. current for the maximum fault level at the next substation. The procedure must be started from the furthest substation, and then continued by moving back towards the source. When the characteristics of two relays cross at a particular system fault level, thus making it difficult to obtain correct coordination, it is necessary to set the instantaneous unit of the relay at the substation that is

furthest away from the source to such a value that the relay operates for a slightly lower level of current, thus avoiding loss of coordination. The 25% margin avoids overlapping the downstream instantaneous unit if a considerable DC component is present. In HV systems operating at 220 kV or above, a higher value should be used since the X/R ratio becomes larger, as does the DC component.

(ii) Distribution lines

The setting of the instantaneous element of relays on distribution lines that supply only pole-mounted MV/LV transformers is dealt with differently from the previous case, since these lines are at the end of the MV system. They therefore do not have to fulfil the coordination conditions that have to be met by lines between substations and so one of the following two values can be used to set these units:

- fifty percent of the maximum short-circuit current at the point of connection of the CT supplying the relay;
- between six and ten times the maximum circuit rating.

(iii) Transformer units

The instantaneous units of the overcurrent relays installed on the primary side of the transformers should be set at a value between 125% and 150% of the short-circuit current existing at the busbar on the low-voltage side, referred to as the high-voltage side. This value is higher than those mentioned previously to avoid lack of coordination with the higher currents encountered due to the magnetic inrush current when energising the transformer. If the instantaneous units of the transformer secondary winding overcurrent protection and the feeder relays are subjected to the same short-circuit level, then the transformer instantaneous units need to be overridden to avoid loss of selectivity unless there are communication links between these units that can permit the disabling of the transformer instantaneous overcurrent protection for faults detected by the feeder instantaneous overcurrent protection.

5.3.2 Coverage of instantaneous units protecting lines between substations

The percentage of coverage of an instantaneous unit that protects a line, X, can be illustrated by considering the system shown in Figure 5.4.

The following parameters are defined:

$$K_{\mathrm{i}} = \frac{I_{\text{pick-up}}}{I_{\text{end}}}$$

and

$$K_{\mathrm{s}} = \frac{Z_{\text{source}}}{Z_{\text{element}}}$$

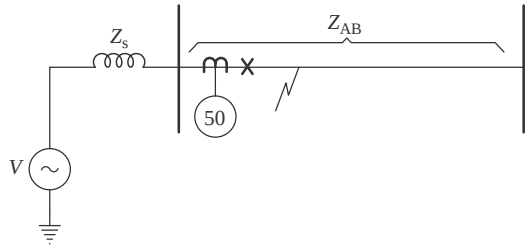


Figure 5.4 Coverage of instantaneous units

From Figure 5.4,

$$I_{\text{pick-up}} = \frac{V}{Z_s + XZ_{ab}} \tag{5.1}$$

where V = voltage at the relay CT point
Z_s = source impedance
Z_{ab} = impedance of the element being protected = Z_{element}
X = percentage of line protected
I_{end} = current at the end of the line
$I_{\text{pick-up}}$ = minimum current value for relay PU

$$I_{\text{end}} = \frac{V}{Z_s + Z_{ab}} \tag{5.2}$$

$$K_i = \frac{Z_s + Z_{ab}}{Z_s + XZ_{ab}} \Rightarrow X = \frac{Z_s + Z_{ab} - Z_s K_i}{Z_{ab} K_i} \tag{5.3}$$

This gives

$$K_s = \frac{Z_s}{Z_{ab}} \Rightarrow X = \frac{K_s(1 - K_i) + 1}{K_i} \tag{5.4}$$

For example, if $K_i = 1.25$ and $K_s = 1$, then $X = 0.6$, i.e. the protection covers 60% of the line.

Example 5.1: The effect of reducing the source impedance, Z_s, on the coverage provided by the instantaneous protection can be appreciated by considering the system in Figure 5.5, and using a value of 1.25 for K_i in (5.4). From this, we get:

Z_s (Ω)	Z_{AB} (Ω)	I_A (A)	I_B (A)	% Coverage
10	10	100	50	60
2	10	500	83	76

5.3.3 Setting the parameters of time-delay overcurrent relays

The operating time of an overcurrent relay has to be delayed to ensure that, in the presence of a fault, the relay does not trip before any other protection situated

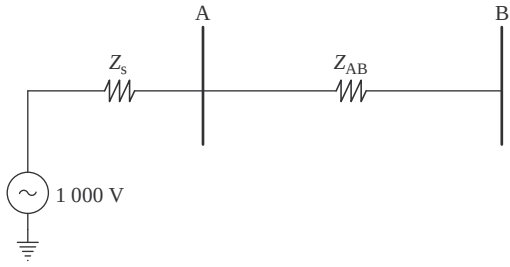


Figure 5.5 Equivalent circuit for Example 5.1

closer to the fault. The curves of inverse-time overcurrent relays associated with two breakers on the same feeder in a typical system are shown in Figure 5.6, illustrating the difference in the operating time of these relays at the same fault levels in order to satisfy the discrimination margin requirements. Definite-time relays and inverse-time relays can be adjusted by selecting two parameters – the time dial or time multiplier setting, and the PU or plug setting (tap setting).

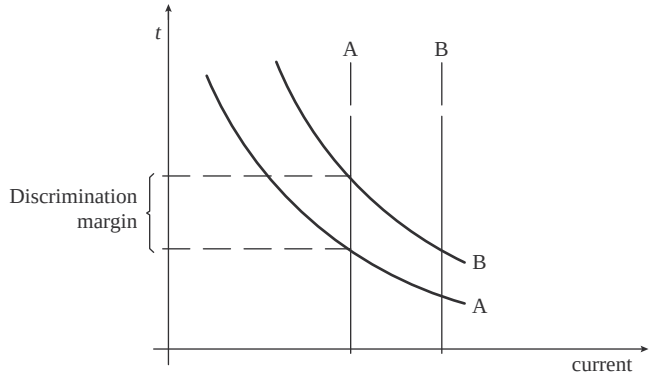


Figure 5.6 Overcurrent inverse-time relay curves associated with two breakers on the same feeder

5.3.3.1 The pick-up setting

The PU setting, or plug setting, is used to define the PU current of the relay, and fault currents seen by the relay are expressed as multiples of this. This value is usually referred to as the plug setting multiplier (PSM), which is defined as the ratio of the fault current in secondary amps to the relay PU or plug setting. For phase relays, the PU setting is determined by allowing a margin for overload above the nominal current, as in the following expression:

$$\text{Pick-up setting} = \frac{\text{OLF} \times I_{\text{nom}}}{\text{CTR}} \tag{5.5}$$

where

OLF = overload factor that depends on the element being protected
I_{nom} = nominal circuit current rating
CTR = CT ratio

The overload factor recommended for motors is 1.05. For lines, transformers and generators, it is normally in the range of 1.25–1.5. In distribution systems where it is possible to increase the loading on feeders under emergency conditions, the overload factor can be of the order of 2. In any case, I_{nom} has to be smaller than those of the CT and the thermal capacity of the conductor; otherwise, the smallest value has to be taken to calculate the PU setting.

For earth-fault relays, the PU setting is determined taking account of the maximum unbalance that would exist in the system under normal operating conditions. A typical unbalance allowance is 20% so that the expression in (5.5) becomes

$$\text{Pick-up setting} = \frac{0.2 \times I_{nom}}{\text{CTR}} \tag{5.6}$$

In HV transmission lines, the unbalance allowance could go down to 10%, whereas in rural distribution feeders the value could be as high as 30%.

5.3.3.2 Time dial setting

The time dial setting adjusts the time delay before the relay operates whenever the fault current reaches a value equal to, or greater than, the relay current setting. In electromechanical relays, the time delay is usually achieved by adjusting the physical distance between the moving and fixed contacts; a smaller time dial value results in shorter operating times. The time dial setting is also referred to as the time multiplier setting.

The criteria and procedures for calculating the time dial setting, to obtain the appropriate protection and coordination for the system, are considered next. These criteria are mainly applicable to inverse-time relays, although the same methodology is valid for definite-time relays.

1. Determine the required operating time t_1 of the relay furthest away from the source by using the lowest time dial setting and considering the fault level for which the instantaneous unit of this relay picks up. This time dial setting may have to be higher if the load that flows when the circuit is re-energised after a loss of supply is high (the cold load PU), or if it is necessary to coordinate with devices installed downstream, e.g. fuses or reclosers.
2. Determine the operating time of the relay associated with the breaker in the next substation towards the source, $t_{2a} = t_1 + t_{margin}$, where t_{2a} is the operating time of the back-up relay associated with breaker 2 and t_{margin} is the discrimination margin. The fault level used for this calculation is the same as that used to determine the timing t_1 of the relay associated with the previous breaker.

3. With the same fault current as in 1 and 2 above, and knowing t_{2a} and the PU value for relay 2, calculate the time dial setting for relay 2. Use the closest available relay time dial setting whose characteristic is above the calculated value.
4. Determine the operating time (t_{2b}) of relay 2, but now using the fault level just before the operation of its instantaneous unit.
5. Continue with the sequence, starting from the second stage.

The procedure referred to above is appropriate if it can be assumed that the relays have their characteristic curves scaled in seconds. For those relays where the time adjustment is given as a percentage of the operating curve for 1 s, the time dial setting can be determined starting from the fastest multiplier applied to the curve for time dial 1. In most modern relays, the time settings can start from values as low as 0.1 s, in steps of 0.1 s.

5.3.3.3 Time discrimination margin

A time discrimination margin between two successive time/current characteristics of the order of 0.25–0.4 s should be typically used. This value avoids losing selectivity due to one or more of the following:

- breaker opening time;
- relay overrun time after the fault has been cleared;
- variations in fault levels, deviations from the characteristic curves of the relays (e.g. due to manufacturing tolerances) and errors in the current transformers.

In numerical relays, there is no overrun, and therefore the margin could be chosen as low as 0.2 s.

Single-phase faults on the star side of a Dy transformer are not seen on the delta side. Therefore, when setting earth-fault relays, the lowest available time dial setting can be applied to the relays on the delta side, which makes it possible to considerably reduce the settings and thus the operating times of earth-fault relays nearer the source infeed.

5.3.3.4 Use of mathematical expressions for the relay characteristics

The procedure indicated above for phase and earth units can easily be used when the operating characteristics of the relays are defined by mathematical formulae instead of by curves on log–log paper. IEC and ANSI/IEEE Standards define the operating time mathematically by the following expression:

$$t = \frac{k\beta}{(I/I_s)^{\alpha} - 1} + L \tag{5.7}$$

where

t = relay operating time in seconds
k = time dial, or time multiplier, setting
I = fault current level in secondary amps

I_s = PU current selected
L = constant

The constants α and β determine the slope of the relay characteristics. The values of α, β and L for various standard overcurrent relay types manufactured under ANSI/IEE and IEC Standards are given in Table 5.1. Typical characteristics for both types are shown in Figures 5.7 and 5.8.

Table 5.1 ANSI/IEEE and IEC constants for standard overcurrent relays

Curve description	**Standard**	α	β	L
Moderately inverse	IEEE	0.02	0.0515	0.114
Very inverse	IEEE	2.0	19.61	0.491
Extremely inverse	IEEE	2.0	28.2	0.1217
Inverse	CO8	2.0	5.95	0.18
Short-time inverse	CO2	0.02	0.0239	0.0169
Standard inverse	IEC	0.02	0.14	0
Very inverse	IEC	1.0	13.5	0
Extremely inverse	IEC	2.0	80.0	0
Long-time inverse	UK	1.0	120	0

Given the relay characteristic, it is a straightforward task to calculate the time response for a given time dial setting k, PU setting and the other values of the expression in (5.7). Likewise, if a particular time response and PU setting have been determined, the time dial setting is found by solving k from the same equation.

5.4 Constraints of relay coordination

5.4.1 Minimum short-circuit levels

When the time delay unit has been set, using maximum fault levels, it is necessary to check that the relays will operate at the minimum fault levels, and in the correct sequence. For this, it is sufficient to verify that the plug setting multiplier – (I/I_s) in (5.7) – under these conditions is greater than 1.5.

5.4.2 Thermal limits

Once the curves for the overcurrent relays have been defined, a check should be made to ensure that they lie below the curves for the designated thermal capacity of machines and cables. In the case of conductors, manufacturers' graphs, which indicate the length of time that different sizes can withstand various short-circuit values, should be used. A typical graph for copper conductors with thermoplastic insulation is given in Figure 5.9. For motors, the manufacturers' information should also be consulted.

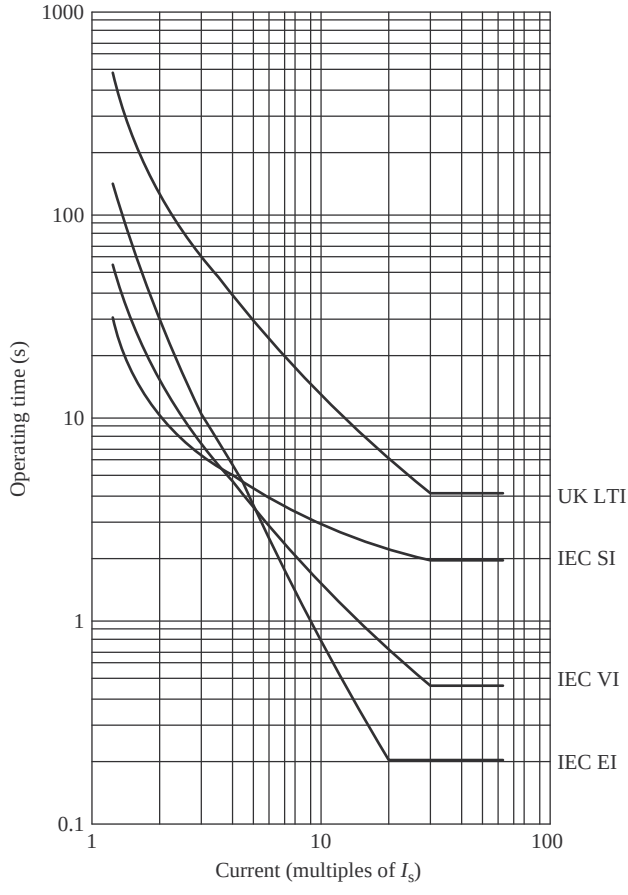


Figure 5.7 Typical IEC overcurrent relay curves

In the case of transformers, the magnitude of the fault current that they can withstand during a given time is limited by their impedance. ANSI/IEEE Standard 242-1986 defines curves of short-circuit capacity for four categories of liquid-immersed transformers, based on the nominal kVA rating of the transformer and the short-circuit impedance.

Figures 5.10–5.13 show the curves of thermal capacity of transformers with the following characteristics:

- Category I – power rating between 5 and 500 kVA single phase; 15–500 kVA three phase.
- Category II – power rating between 501 and 1 667 kVA single phase; 501–5 000 kVA three phase.
- Category III – power rating between 1 668 and 10 000 kVA single phase; 5 001–30 000 kVA three phase.

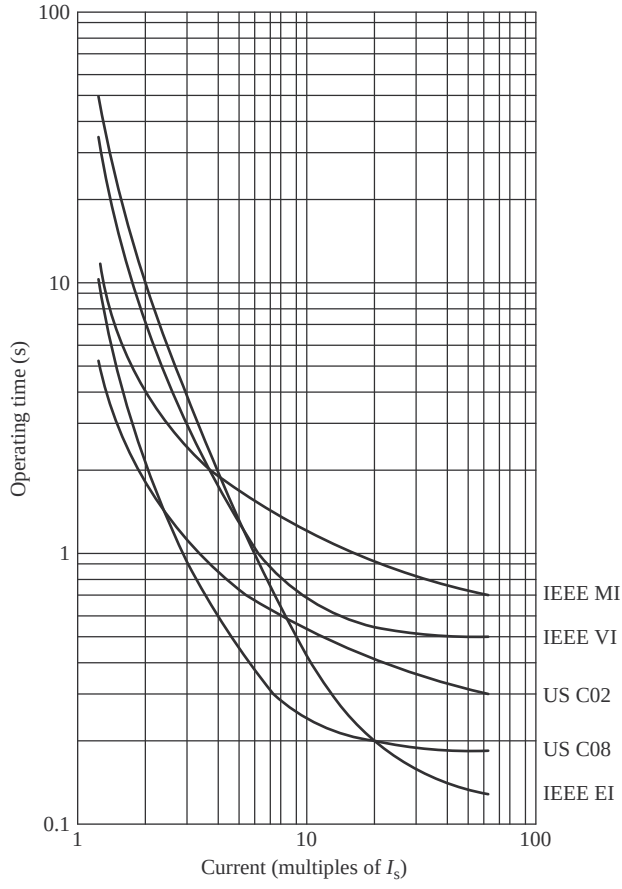


Figure 5.8 Typical ANSI/IEEE overcurrent relay curves

- Category IV – power rating above 10 000 kVA single phase; above 30 000 kVA three phase.

The thermal limit curves for Dy transformers have to be shifted to the left by a ratio of $1/\sqrt{3}$ to make them more sensitive. This compensates for the lower value of current seen by the relays installed on the primary side, relative to the currents seen by the relays on the secondary side, during single-phase fault conditions, as discussed in section 5.5.

5.4.3 Pick-up values

It is also important to check that the relay settings are not going to present problems when other system elements are energised. This is particularly critical for motors, and the appropriate code letter, which indicates the number of times nominal current taken when the motor is starting, should always be borne in mind.

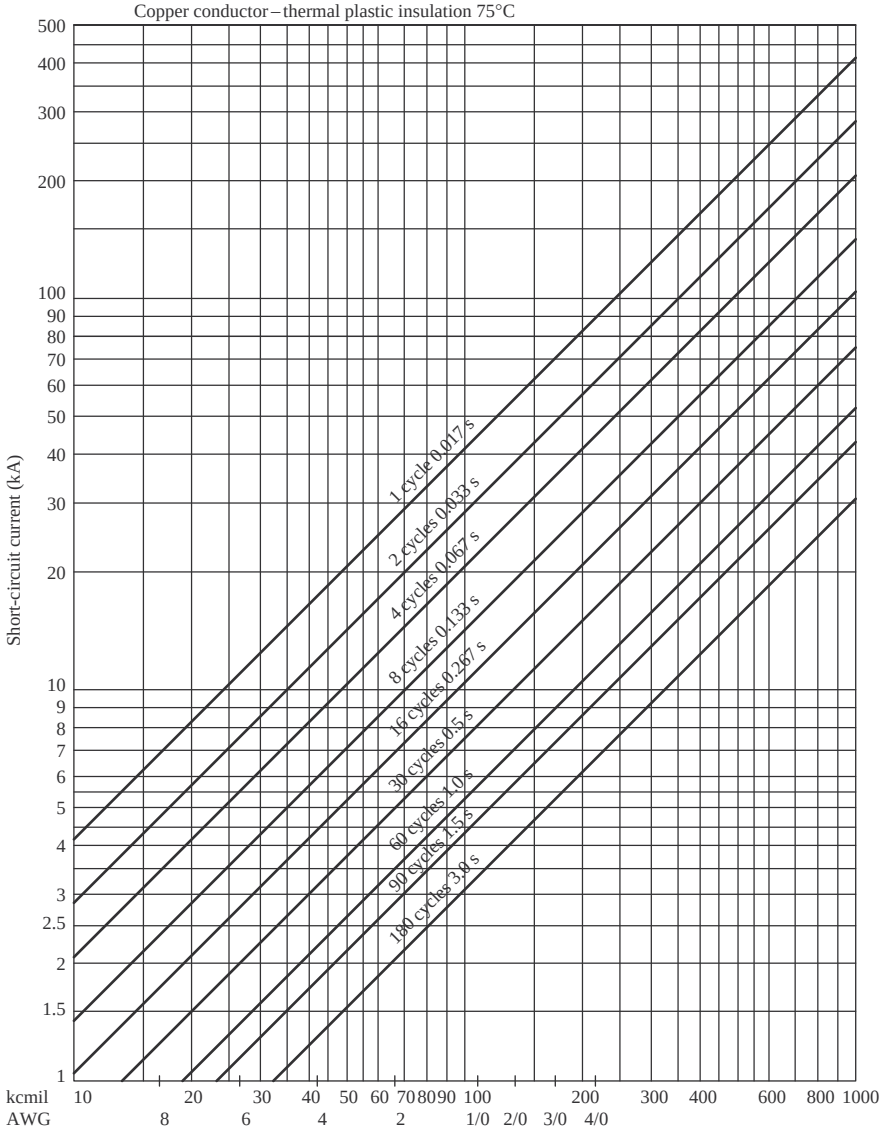


Figure 5.9 Thermal limits of copper conductors with thermoplastic insulation at 75° conductor temperature

In the case of transformers, the initial magnetisation inrush current that a transformer takes can be expressed as $I_{\text{Inrush}} = K \times I_{\text{nom}}$, where I_{nom} is the nominal transformer current and the constant K depends on the transformer capacity; from 500 to 2 500 kVA $K = 8$, and above 2 500 kVA $K = 10$. The inrush point then remains defined by the appropriate inrush current during 0.1 s.

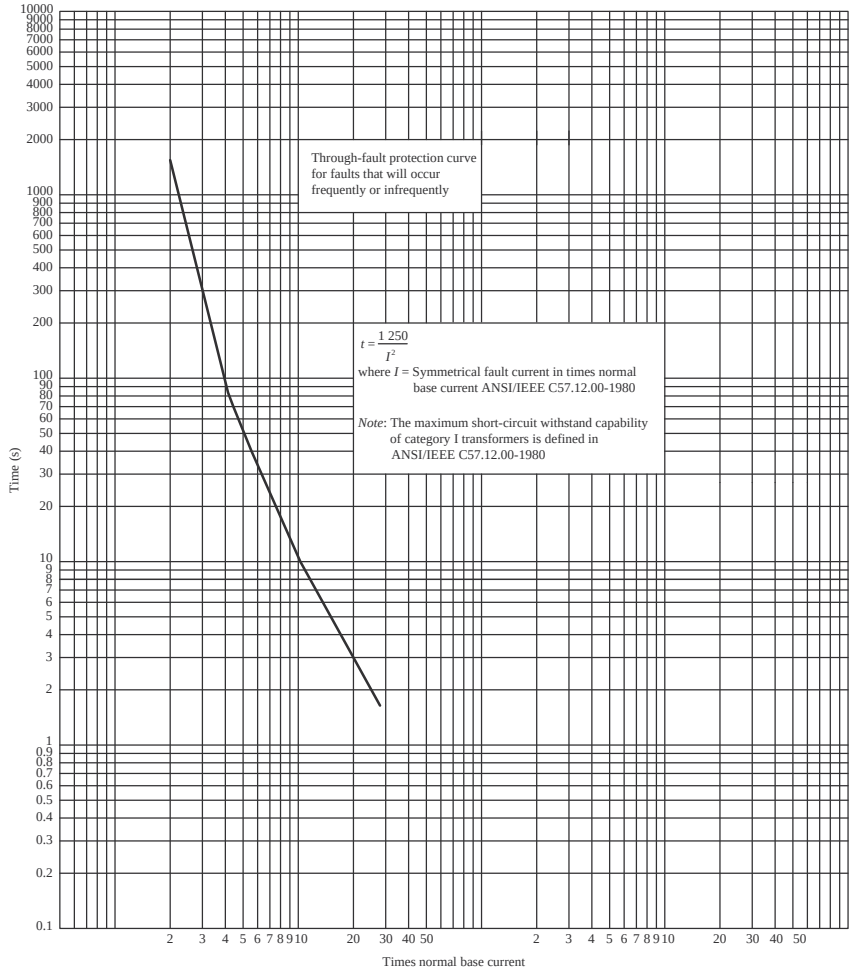


Figure 5.10 Thermal capacity of transformers between 5 and 500 kVA single phase; 15–500 kVA three phase [from ANSI/IEEE Standard 242-1986; reproduced by permission of the IEEE]

Example 5.2: For the system shown in Figure 5.14, and starting from the data that are given there, carry out the following:

- Calculate the nominal currents and three-phase short-circuit levels at each breaker.
- Select the transformation ratios of the CTs.
- Determine the values of the PU setting, time dial and instantaneous settings of all phase relays to ensure a coordinated protection arrangement.
- Find the percentage of the line BC protected by the instantaneous unit of the overcurrent relay associated with breaker 2.
- Draw the time/current characteristics of the relays on the system.

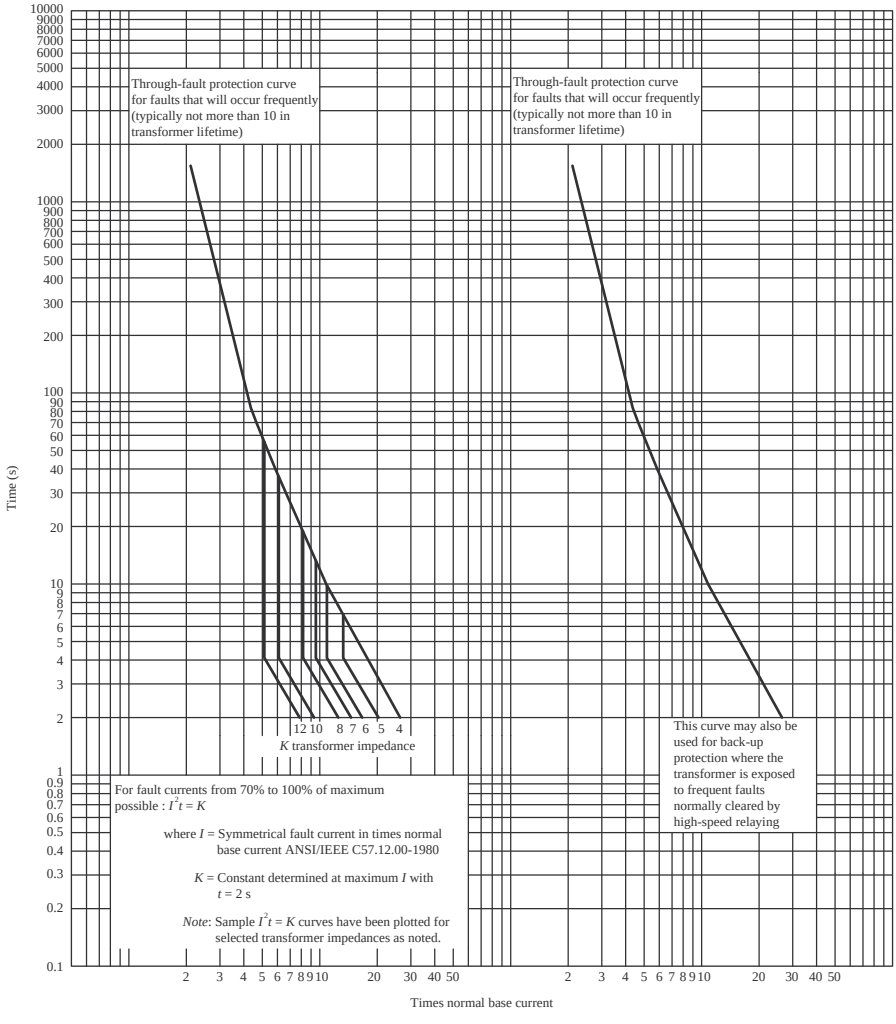


Figure 5.11 Thermal capacity of transformers between 501 and 1 667 single phase; 501–5 000 kVA three phase [from ANSI/IEEE Standard 242-1986; reproduced by permission of the IEEE]

Take into account the following considerations:

- The discrimination margin to be 0.4 s.
- All relays have inverse time characteristics, as shown in Figure 5.15.
- Relay data:

  PU setting: 1–12 A in steps of 1 A;
  time dial setting: as in Figure 5.15;
  instantaneous: 6–144 A in steps of 1 A.

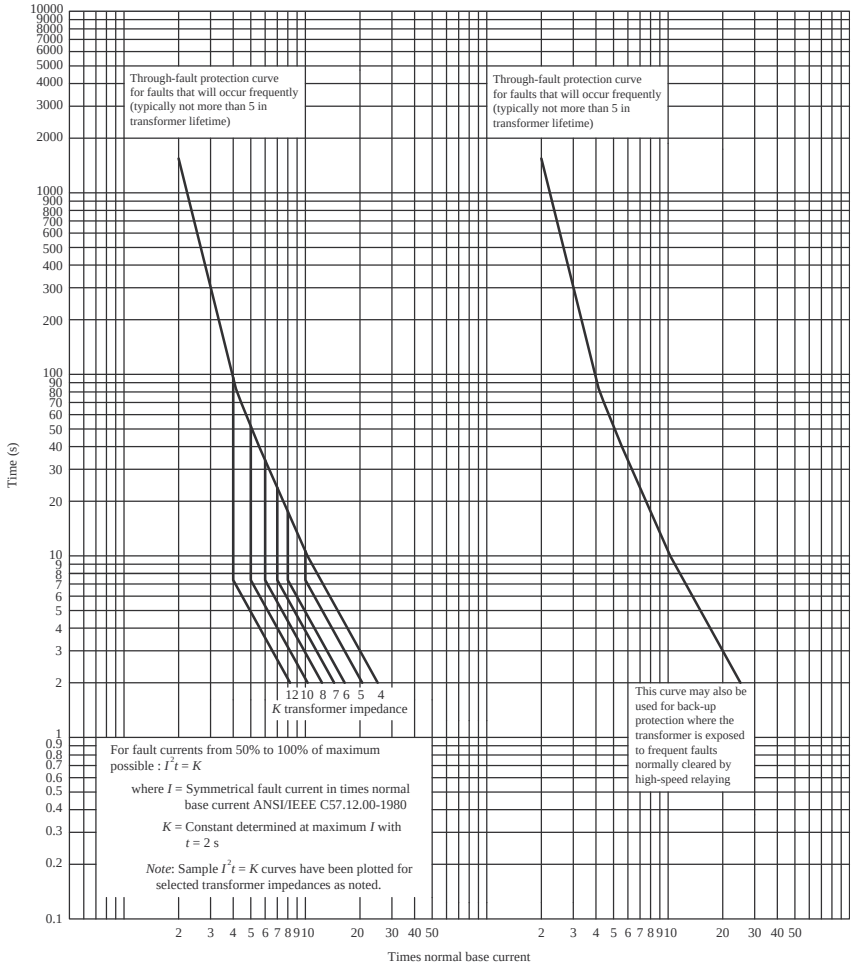


Figure 5.12 Thermal capacity of transformers between 1 668 and 10 000 kVA single-phase; 5 001–30 000 kVA three phase [from ANSI/IEEE Standard 242-1986; reproduced by permission of the IEEE]

Solution

Calculation of nominal currents and three-phase short-circuit levels: From Figure 5.14, the short-circuit level at busbar A, and the impedance of the line BC, can be obtained:

$$Z_{\text{source}} = \frac{V^2}{P_{\text{sc}}} = \frac{(115 + 10^3)^2}{950 \times 10^6} = 13.92\,\Omega, \text{ referred to } 115\,\text{kV}$$

$$Z_{\text{transf}} = Z_{\text{pu}} \times Z_{\text{Base}} = 0.048 \times \frac{(115 \times 10^3)^2}{25 \times 10^6} = 25.39\,\Omega, \text{ referred to } 115\,\text{kV}$$

$$Z_{\text{lineBC}} = 85.35\,\Omega, \text{ referred to } 115\,\text{kV}$$

The equivalent circuit of the system referred to 115 kV is shown in Figure 5.16.

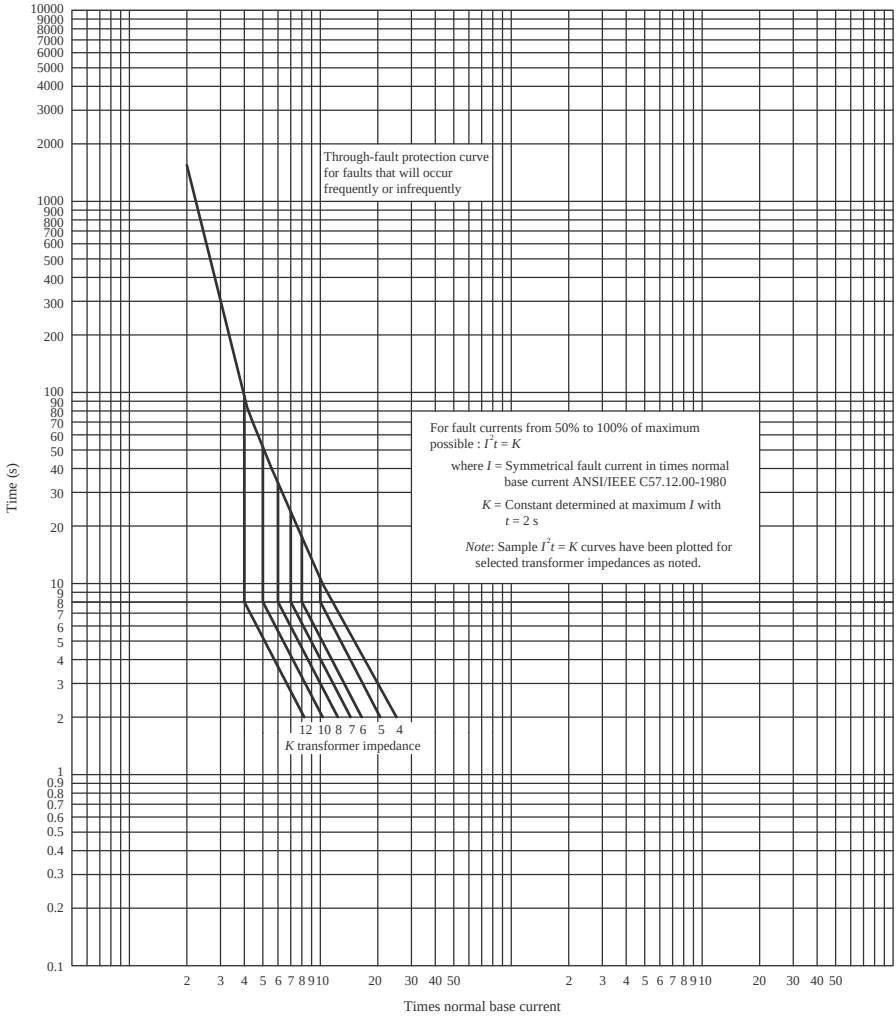


Figure 5.13 Thermal capacity of transformers above 10 000 kVA single phase; 30 000 kVA three phase [from ANSI/IEEE Standard 242-1986; reproduced by permission of the IEEE]

Nominal currents

$$I_{\text{nom1}} = \frac{P}{\sqrt{3} \times V} = \frac{3 \times 10^6}{\sqrt{3}(13.2 \times 10^3)} = 131.2 \text{ A}$$

$$I_{\text{nom2}} = 3 \times I_{\text{nom1}} = 3 \times 131.2 = 393.6 \text{ A}$$

$$I_{\text{nom3}} = \frac{25 \times 10^6}{\sqrt{3}(13.2 \times 10^3)} = 1\,093.5 \text{ A}$$

$$I_{\text{nom4}} = \frac{25 \times 10^6}{\sqrt{3}(115 \times 10^3)} = I_{\text{nom3}} \times \left(\frac{13.2}{115}\right) = 125.5 \text{ A}$$

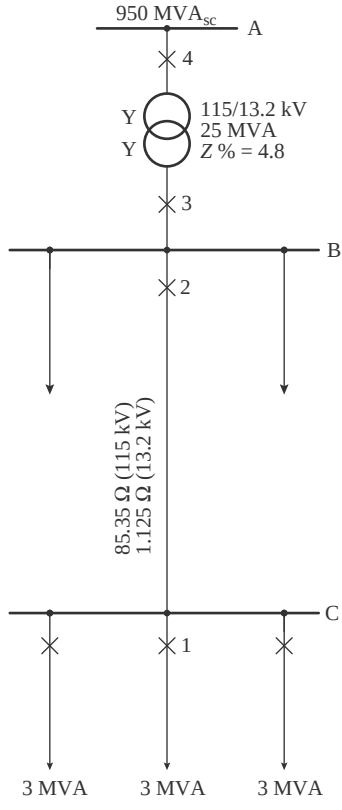


Figure 5.14 Schematic diagram for Example 5.2

Short-circuit levels
The equivalent circuit gives:

$$I_{\text{faultC}} = \frac{115 \times 10^3}{\sqrt{3}(13.92 + 25.39 + 83.35)} = 532.6\,\text{A, referred to 115 kV}$$

$$= 532.6\left(\frac{115}{13.2}\right) = 4\,640.2\,\text{A, referred to 13.2 kV}$$

$$I_{\text{faultB}} = \frac{115 \times 10^3}{\sqrt{3}(13.92 + 25.39)} = 1\,689.0\,\text{A, referred to 115 kV}$$

$$= 1\,689\left(\frac{115}{13.2}\right) = 14\,714.8\,\text{A, referred to 13.2 kV}$$

$$I_{\text{faultA}} = \frac{115 \times 10^3}{\sqrt{3} \times 13.92} = 4\,769.8\,\text{A, referred to 115 kV}$$

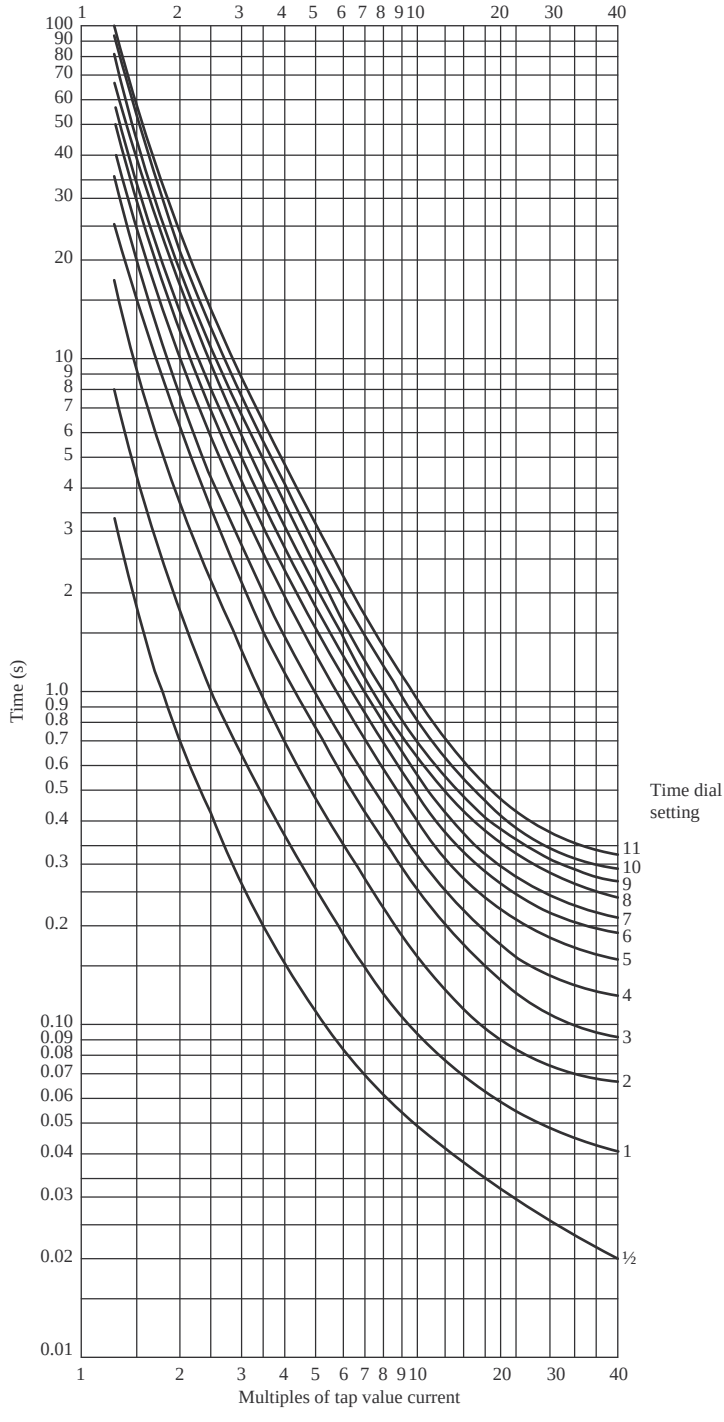


Figure 5.15 Typical operating curves for an inverse-time relay

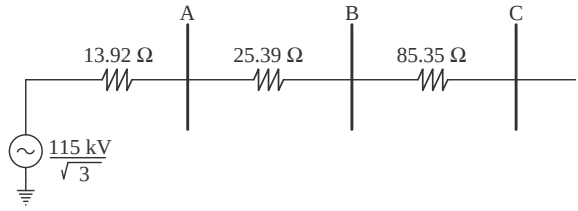


Figure 5.16 Equivalent circuit of the system shown in Figure 5.14

Choice of CT transformer ratio
The transformation ratio of the CTs is determined by the larger of the following two values:

- I_{nom}
- maximum short-circuit current without saturation being present. To fulfil this condition and assuming that a C100 core is used and that the total burden is 1 Ω, $I_{sc}(5/X) \leq 100$ A, where I_{sc} is the short-circuit current.

Table 5.2 summarises the calculations.

Table 5.2 Nominal currents, short-circuit currents and CT ratios for Example 5.2

Breaker number	P_{nom} (MVA)	I_{nom} (A)	I_{sc} (A)	(5/100) I_{sc} (A)	CT ratio
1	3	131.2	4 640.0	232.0	300/5
2	9	393.6	14 714.8	1 735.7	800/5
3	25	1 093.5	14 714.8	735.7	1 100/5
4	25	125.5	4 769.8	238.5	300/5

Determination of the PU setting, time dial and instantaneous setting values: calculation of the PU settings

Relay 1: $1.5(131.2)5/300 = 3.28$ A; ⇒ PU set at 4 A
Relay 2: $1.5(393.6)5/800 = 3.68$ A; ⇒ PU set at 4 A
Relay 3: $1.5(1093.5)5/1100 = 7.46$ A; ⇒ PU set at 8 A
Relay 4: $1.5(125.5)5/300 = 3.14$ A; ⇒ PU set at 4 A

Determination of time dial setting and calibration of the instantaneous setting
Relay 1:

$$I_{\text{pick-up}} = \frac{4 \times 300}{5} = 240 \text{ A}$$

Time dial setting selected is 1.0.

Setting of instantaneous element $= (0.5I_{sc})(1/\text{CTR}) = (0.5 \times 4\,640)5/300 =$ 38.67 A; set at 39 A.

$I_{\text{inst. trip}} = 39 \times 300/5 = 2\,340$ A, primary at 13.2 kV.

Plug setting multiplier, $\text{PSM}_\text{b} = (2\,340 \text{ A} \times 5/300) \times 1/4 = 9.75$ times.

From Figure 5.15, with a plug setting multiplier of 9.75 and a time dial setting at 1, $t_{1\text{b}} = 0.1$ s.

Relay 2:

2 340 A should produce operation of $t_{2\text{a}}$ in at least $0.1 + 0.4 = 0.5$ s.

$$\text{PSM}_\text{a} = 2\,340\,\text{A} \times \frac{5}{800} \times \frac{1}{4} = 3.66 \text{ times}$$

With PSM at 3.66 times, and t_{op} at least 0.5 s, time dial = 2 is chosen.

Instantaneous setting $= (1.25I_{\text{faultC}})(1/\text{CTR}) = 1.25(4\,640)(5/800) = 36.25$ A; set at 37 A.

$$I_{\text{inst.prim}} = (37)\frac{800}{5} = 5\ 920 \text{ A at } 13.2 \text{ kV}$$

$$\text{PSM}_\text{b} = 5\ 920 \text{ A} \times \frac{5}{800} \times \frac{1}{4} = 9.25 \text{ times}$$

With 9.25 PSM and time dial setting $= 2 \Rightarrow t_{2\text{b}} = 0.18$ s.

Relay 3:

To discriminate with relay 2, take $I_{\text{inst.prim2}} = 5\,920$ A.

Require operation of $t_{3\text{a}}$ in at least $0.18 + 0.4 = 0.58$ s.

$$\text{PSM}_\text{a} = 5\ 920 \text{ A} \times \frac{5}{1\ 100} \times \frac{1}{8} = 3.36 \text{ times}$$

With 3.36 PSM and $t_{\text{op}} = 0.58$ s, $\Rightarrow$ time dial setting = 2.

However, the instantaneous element of the relay associated with breaker 3 is overridden and the discrimination time is applied for a fault on busbar B to avoid lack of coordination with the instantaneous units of the relays associated with the feeders from the busbar, as referred to in section 5.3.1.

On the basis of $I_{\text{sc}} = 14\,714.8$ at 13.2 kV,

$$\text{PSM}_\text{b} = 14\ 714.8\,\text{A} \times \frac{5}{1\,100} \times \frac{1}{8} = 8.36 \text{ times}$$

With 8.36 times PSM and time dial setting = 2, $\Rightarrow t_{3\text{b}} = 0.21$ s.

Relay 4:

For 14 714.8 A, PSM $= 14\,714.8$ A$(13.2/115)5/300 \times 1/4 = 7.04$ times.

Require $t_4 = 0.21 + 0.4 = 0.61$ s.

With 7.04 PSM and $t_{\text{op}} = 0.61$ s $\Rightarrow$ time dial setting = 5.

$$\text{Setting of instantaneous element} = (1.25 \times I_{\text{faultB}})\left(\frac{1}{\text{CTR}}\right)$$
$$= 1.25(1\,689)\left(\frac{5}{300}\right) = 35.19\,\text{A}.$$

Setting = 36 A.

$$I_{\text{inst.prim}} = 36\left(\frac{300}{5}\right) = 2\,160 \text{ A, referred to 115 kV.}$$

$$I_{\text{inst.prim}} = 2\,160\left(\frac{115}{13.2}\right) = 18\,818.2 \text{ A, referred to 13.2 kV.}$$

Table 5.3 summarises the four relay settings.

Table 5.3 Summary of settings for Example 5.2

Relay associated with breaker number	**Pick-up (A)**	**Time dial**	**Instantaneous I_{sec} (A)**	**Instantaneous I_{prim} (A)**
1	2.5	–	40	2 400
2	4.0	2	37	5 920
3	8.0	2	–	–
4	4.0	5	36	18 818

Percentage of line B–C protected by the instantaneous element of the relay associated with breaker 2:

$$X\,(\%) = \frac{K_{\text{s}}(1 - K_{\text{i}}) + 1}{K_{\text{i}}}$$

$$K_{\text{i}} = \frac{I_{\text{sc pick-up}}}{I_{\text{sc end}}} = \frac{5\,920}{4\,640} = 1.28$$

$$K_{\text{s}} = \frac{Z_{\text{source}}}{Z_{\text{element}}} = \frac{13.92 + 25.39}{85.35} = 0.46$$

and

$$X\,(\%) = \frac{0.46(1 - 1.28) + 1}{1.28} = 0.68$$

Therefore, the instantaneous element covers 68% of the line BC.

The coordination curves of the relays associated with this system are shown in Figure 5.17. It should be noted that these are all drawn for currents at the same voltage – in this case 13.2 kV.

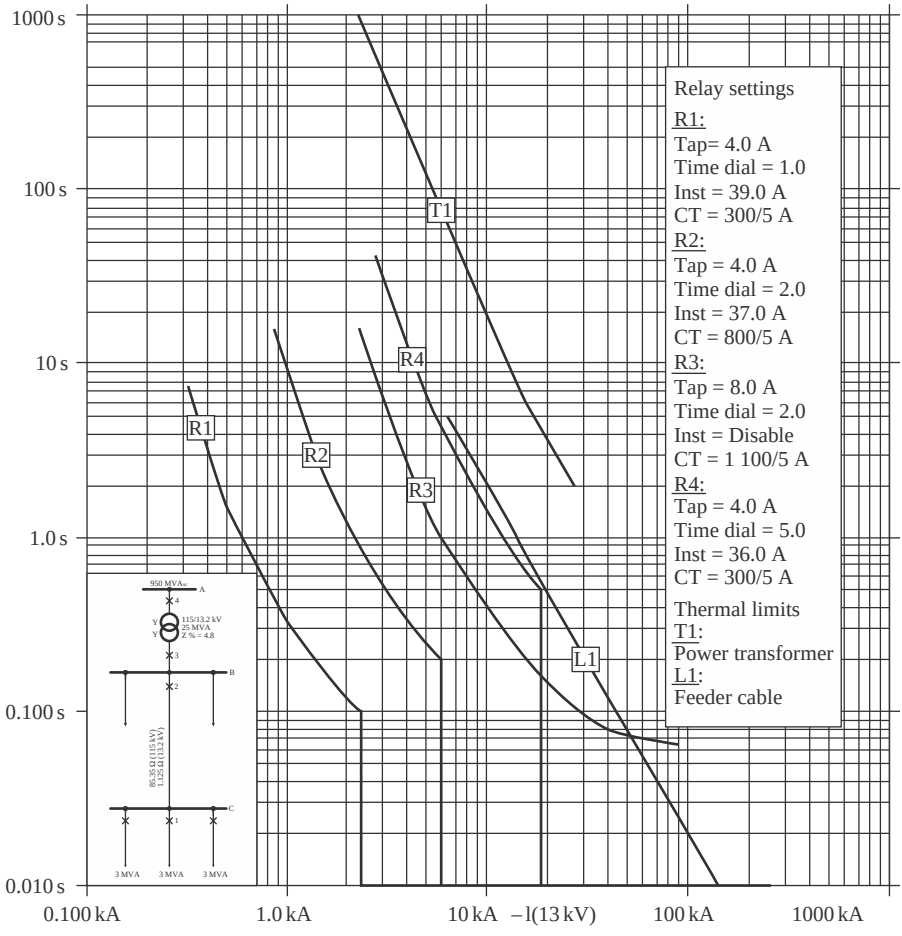


Figure 5.17 Relay coordination curves for Example 5.2

5.5 Coordination across Dy transformers

In the case of overcurrent relay coordination for Dy transformers, the distribution of currents in these transformers should be checked for three-phase, phase-to-phase and single-phase faults on the secondary winding, shown in Figure 5.18.

To simplify the operations, it can be assumed that the voltages between the phases of the transformer are the same, for both the primary and the secondary windings. Thus, the number of turns on the primary is equal to $\sqrt{3}$ times the number of turns of the secondary, i.e. $N_1 = \sqrt{3}N_2$.

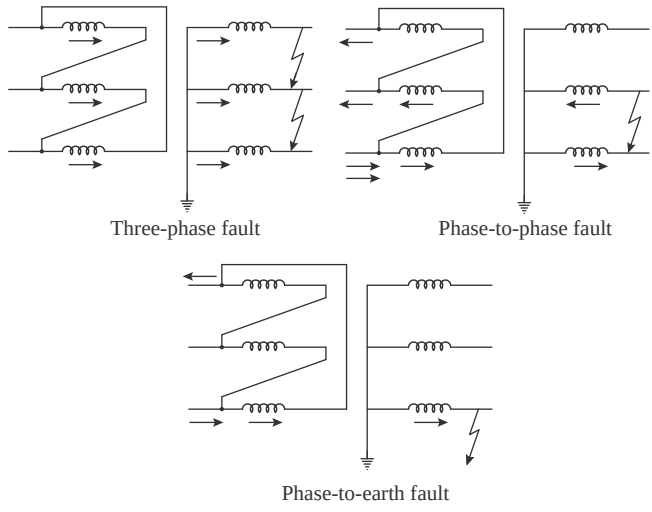


Figure 5.18 Distribution of current for a fault on a Dy transformer

5.5.1 Three-phase fault

$$I_f = \frac{E_{\phi-\mathrm{n}}}{X} = I \tag{5.8}$$

$$I_{\text{delta}} = I\frac{N_2}{N_1} = \frac{I}{\sqrt{3}} \tag{5.9}$$

$$I_{\text{primary}} = \sqrt{3}I_{\text{delta}} = I \tag{5.10}$$

From the above, it can be seen that the currents that flow through the relays associated with the secondary winding are equal to the currents that flow through those relays associated with the primary winding, as expected, since the primary and secondary voltages are equal and the fault involves all three phases.

5.5.2 Phase-to-phase fault

$$I_{\mathrm{f}} = \frac{E_{\phi-\phi}}{2X} = \frac{\sqrt{3}\times E_{\phi-\phi}}{2X} = \frac{\sqrt{3}}{2}I \tag{5.11}$$

$$I_{\text{delta}} = \frac{\sqrt{3}}{2}\times I \times \frac{N_2}{N_1} = \frac{I}{2} \tag{5.12}$$

$$I_{\text{primary}} = 2I_{\text{delta}} = I \tag{5.13}$$

For this case, the current that goes through the relays installed in the secondary winding circuit is equal to $\sqrt{3}/2$ times the current that flows through the relays

associated with the primary on the phase that has the largest current value. From Figure 5.18, it is clear that, for this fault, the current distribution at the primary is 1–1–2, and 0–1–1 at the secondary.

5.5.3 Phase-to-earth fault

$$I_{\mathrm{f}} = \frac{E_{\phi-\mathrm{n}}}{X} = I \tag{5.14}$$

$$I_{\mathrm{delta}} = I \times \frac{N_2}{N_1} = \frac{I}{\sqrt{3}} \tag{5.15}$$

$$I_{\mathrm{primary}} = \frac{I}{\sqrt{3}} \tag{5.16}$$

Thus, for a phase-to-earth fault, the current through the relays installed in the secondary winding circuit on the faulted phase is equal to $\sqrt{3}$ times the current that flows through the relays associated with the primary winding on the same phase.

The results of the three cases are summarised in Table 5.4. Analysing the results, it can be seen that the critical case for the coordination of overcurrent relays is the phase-to-phase fault. In this case, the relays installed in the secondary carry a current less than the equivalent current flowing through the primary relays that could lead to a situation where the selectivity between the two relays is at risk. For this reason, the discrimination margin between the relays is based on the operating time of the secondary relays at a current equal to $\sqrt{3}I_{\mathrm{f}}/2$, and on the operating time for the primary relays for the full fault current value I_{f}, as shown in Figure 5.19.

Table 5.4 Summary of fault conditions

Fault	I_{primary}	$I_{\mathrm{secondary}}$
Three-phase	I	I
Phase-to-phase	I	$\sqrt{3}I/2$
Phase-to-earth	I	$\sqrt{3}I$

Example 5.3: For the system shown in Figure 5.20, calculate the following:

- The three-phase short-circuit levels on busbars 1 and 2.
- The transformation ratios of the CTs associated with breakers 1–8, given that the primary turns are multiples of 100 except for the CT for breaker number 9 that has a ratio of 250/5. Assume that the total burden connected to each CT is 1 Ω and that C100 cores are used.
- The settings of the instantaneous elements, and the PU and time dial settings of the relays to guarantee a coordinated protection arrangement, allowing a discrimination margin of 0.4 s.

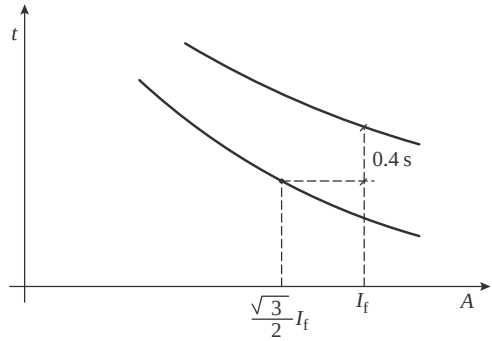


Figure 5.19 Coordination of overcurrent relays for a Dy transformer

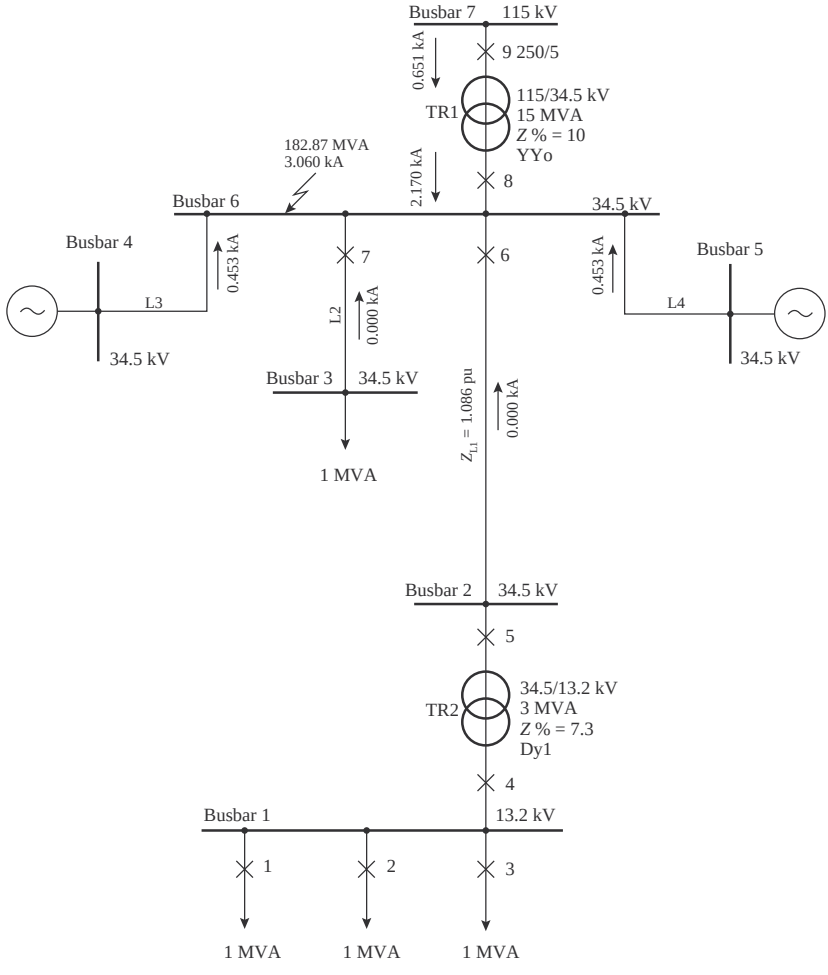


Figure 5.20 Single-line diagram for Example 5.3 showing fault currents

- The percentage of the 34.5-kV line protected by the instantaneous element of the overcurrent relay associated with breaker 6.

Take into account the following additional information:

- The PU impedances are calculated on the following bases:
  $V = 34.5$ kV
  $P = 100$ MVA
- The settings of relay 7 are:
  PU = 4 A
  Time dial = 0.3
  Instantaneous = 1 100 A primary current
- All the relays have an IEC very inverse-time characteristic with the values given in Table 5.1.
- Relay data:
  PU setting = 1–12 A in steps of 1 A
  Time dial setting = 0.05–10 in steps of 0.05
  Instantaneous: 6–14 A in steps of 1 A
- The setting of the instantaneous elements of the relays associated with the feeders is to be carried out on the basis of ten times the maximum nominal current.
- The short-circuit MVA and fault currents, for a fault on the 34.5-kV busbar at substation A, are given in Figure 5.20.

Solution

Calculation of equivalent impedance: The short-circuit level on the 34.5-kV busbar at substation A can be obtained from the values in Figure 5.20 (183.11 MVA). Using this, the equivalent impedance of the system behind the busbar is calculated as follows:

$$Z_{\text{base}} = \frac{V^2}{P_{\text{sc}}} = \frac{(34\,500)^2}{183.11 \times 10^6} = 6.5\,\Omega, \text{ referred to } 34.5\,\text{kV}$$

$$Z_{\text{transf 1}} = 0.1\,\frac{(34\,500)^2}{15 \times 10^6} = 7.93\,\Omega, \text{ referred to } 34.5\,\text{kV}$$

$$= 88.17 \text{ W, referred to } 115\,\text{kV}$$

$$Z_{\text{transf 2}} = 0.073\,\frac{(34\,500)^2}{3 \times 10^6} = 28.96\,\Omega, \text{ referred to } 34.5\,\text{kV}$$

$$Z_{\text{line}} = 1.086\,\frac{(34\,500)^2}{100 \times 10^6} = 12.93\,\Omega, \text{ referred to } 34.5\,\text{kV}$$

The equivalent positive-sequence network, referred to 34.5 kV, is shown in Figure 5.21.

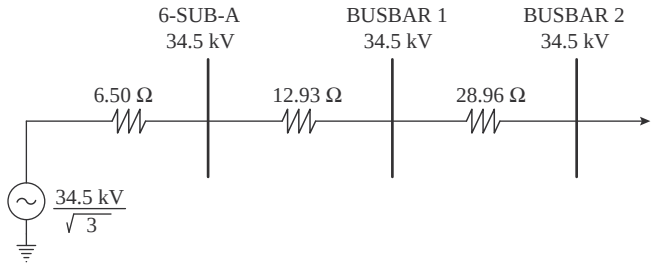


Figure 5.21 Positive-sequence network for Example 5.3

Nominal currents

$$I_{\text{nom1,2,3}} = \frac{1 \times 10^6}{\sqrt{3} \times 13.2 \times 10^3} = 43.74\,\text{A} \quad \text{at } 13.2\,\text{kV}$$

$$I_{\text{nom4}} = \frac{3 \times 10^6}{\sqrt{3} \times 13.2 \times 10^3} = 131.22\,\text{A} \quad \text{at } 13.2\,\text{kV}$$

$$I_{\text{nom5}} = \frac{3 \times 10^6}{\sqrt{3} \times 34.5 \times 10^3} = 50.20\,\text{A} \quad \text{at } 34.5\,\text{kV}$$

$$I_{\text{nom6}} = 50.20\,\text{A} \quad \text{at } 34.5\,\text{kV}$$

$$I_{\text{nom7}} = \frac{1 \times 10^6}{\sqrt{3} \times 34.5 \times 10^3} = 16.73\,\text{A}$$

$$I_{\text{nom8}} = 251.02\,\text{A} \quad \text{at } 34.5\,\text{kV}$$

$$I_{\text{nom9}} = \frac{15 \times 10^6}{\sqrt{3} \times 115 \times 10^3} = 75.31\,\text{A} \quad \text{at } 115\,\text{kV}$$

Short-circuit levels

The short-circuit MVA (P_{sc}) of the transformer at 34.5 kV $= \sqrt{3} \times 2\,170.34 \times 34.5 \times 10^3 = 129.69$ MVA.

$$Z_{\text{transf1}} + Z_{\text{base}} = \frac{(34.5)^2}{129.69} = 9.18\,\Omega, \text{ referred to } 34.5\,\text{kV}$$

$$= 101.97\,\Omega, \text{ referred to } 115\,\text{kV}$$

$$Z_{\text{system}} = 101.97\,\Omega - 88.17\,\Omega = 13.80\,\Omega, \text{ referred to } 115\,\text{kV}$$

$$I_{\text{fault1,2,3,4}} = \frac{34.5 \times 10^3}{\sqrt{3}(6.5 + 12.93 + 28.96)} = 411.63\,\text{A} \quad \text{at } 34.5\,\text{kV}$$

$$= 1\,075.84\,\text{A} \quad \text{at } 13.2\,\text{kV}$$

$$I_{\text{fault5}} = \frac{34.5 \times 10^3}{\sqrt{3}(6.5 + 12.93)} = 1\,025.15\,\text{A} \quad \text{at } 34.5\,\text{kV}$$

$$I_{\text{fault6,7}} = \frac{34.5 \times 10^3}{\sqrt{3}(6.5)} = 3\,064.40\,\text{A} \quad \text{at } 34.5\,\text{kV}$$

$$I_{\text{fault8}} = \frac{129.69 \times 10^6}{\sqrt{3} \times 34.5 \times 10^3} = 2\,170.34\,\text{A} \quad \text{at } 34.5\,\text{kV}$$

$$I_{\text{fault9}} = \frac{115 \times 10^3}{\sqrt{3} \times 13.80} = 4\,811.25\,\text{A} \quad \text{at } 115\,\text{kV}$$

Selection of current transformers

Table 5.5 gives the main values for determining the transformation ratio of the CTs, which is taken as the larger of the following two values:

- nominal current;
- maximum short-circuit current for which no saturation is present.

Therefore, $(I_{\text{sc}} \times 5/X) \leq 100 \Rightarrow X\ (I_{\text{sc}} \times 5/100)$.

Table 5.5 Determination of CT ratios for Example 5.3

Breaker number	P_{nom} (MVA)	I_{nom} (A)	I_{sc} (A)	(5/100) I_{sc} (A)	CT ratio
9	15	75.31	4 797.35	239.87	250/5
8	15	251.02	2 170.40	108.51	300/5
7	1	16.73	3 060.34	153.01	200/5
6	3	50.20	3 060.34	153.01	200/5
5	3	50.20	1 025.67	51.28	100/5
4	3	131.22	1 076.06	53.80	200/5
1–3	1	43.74	1 076.06	53.80	100/5

Determining the PU values

$$I_{\text{load 1,2,3}} = 43.74\,\text{A};\ \text{PU}_{1,2,3} = (1.5)(43.74)\left(\frac{5}{100}\right) = 3.28;\ \text{PU}_{1,2,3} = 4\,\text{A}$$

$$I_{\text{load 4}} = 131.22\,\text{A};\ \text{PU}_4 = (1.5)(131.22)\left(\frac{5}{200}\right) = 4.92;\ \text{PU}_4 = 5\,\text{A}$$

$$I_{\text{load5}} = 50.20\,\text{A};\ \text{PU}_5 = (1.5)(50.20)\left(\frac{5}{100}\right) = 3.76\,\text{A};\ \text{PU}_5 = 4\,\text{A}$$

$$I_{\text{load6}} = 50.20\,\text{A};\ \text{PU}_6 = (1.5)(50.20)\left(\frac{5}{200}\right) = 1.88\,\text{A};\ \text{PU}_6 = 2\,\text{A}$$

$$\text{PU}_7 = 4\,\text{A}$$

$$I_{\text{load8}} = 251.02\,\text{A};\ \text{PU}_8 = (1.5)(251.02)\left(\frac{5}{300}\right) = 6.28\,\text{A};\ \text{PU}_8 = 7\,\text{A}$$

$$I_{\text{load9}} = 75.31\,\text{A};\ \text{PU}_9 = (1.5)(75.31)\left(\frac{5}{250}\right) = 2.26\,\text{A};\ \text{PU}_9 = 3\,\text{A}$$

Determining the instantaneous and time dial settings

Relays 1–3:

When calculating the settings for the relays situated at the end of the circuit, the minimum time dial setting of 0.05 is selected. From the given information, the setting of the instantaneous element is based on ten times the maximum load current seen by the relay.

Thus,

$$I_{\text{inst. trip}} = 10 \times I_{\text{nom}} \times \left(\frac{1}{\text{CTR}}\right) = 10 \times 43.74 \times \left(\frac{5}{100}\right) = 21.87\,\text{A}$$

$$\Rightarrow \text{set at } 22\,\text{A}$$

$$I_{\text{prim. trip}} = 22\left(\frac{100}{5}\right) = 440\,\text{A}$$

Given the constants for the IEC, very inverse overcurrent relay are $\alpha = 1.0$, $\beta = 13.5$ and $L = 0$; then, from (5.7), the relay operating time t is [(time dial setting) $\times$ 13.5]/(PSM – 1), where PSM is the ratio of the fault current in secondary amps to the relay PU current. PSM = 22/4 = 5.5 times and, with the time dial setting of 0.05, the relay operating time is (0.05 $\times$ 13.5)/(5.5 – 1) = 0.15 s.

Relay 4:

To discriminate with relay 3 at 440 A requires operation in t_{4a} – 0.15 + 0.4 = 0.55 s.

PSM_{4a} = (440 $\times$ 5/200)(1/5) = 2.2 times. At 2.2 times, and t_{4a} = 0.55 s, the time dial setting = 0.55 $\times$ (2.2 – 1)/13.5 = 0.049 $\Rightarrow$ 0.05.

This relay has no setting for the instantaneous element, as referred to in section 5.3.1.

The operating time for a line-to-line fault is determined by taking 86% of the three-phase fault current. PSM_{4b} = (0.86)(1 075.84 $\times$ 5/200)(1/5) = 4.63 times. By similar calculations to those for relays 1–3, t_{4b} = 0.19 s.

Relay 5:
The back-up to relay 4 is obtained by considering the operating time for a line-to-line fault of $t_{5a} = 0.9 + 0.4 = 0.59$ s.

$PSM_{5a} = 1\,075.84 \times (13.2/34.5) \times (5/100) \times (1/4) = 5.15$ times. At 5.15 times, and $t_{5a} = 0.59$ s, this gives a required time dial setting of 0.20.

The setting of the instantaneous element is $1.25(1\,075.84) \times (13.2/34.5) \times (5/100) = 25.73$ A $\Rightarrow$ 26 A, so that $I_{\text{prim. trip}} = 26(100/5) = 520$ A.

The operating time of the time-delay element is calculated from $PSM_{5b} = (1/4) \times 26 = 6.5$ times. At 6.5 times, and with a time dial setting of 0.20, from the relay characteristics and (5.7), $t_{5b} = 0.5$ s.

Relay 6:
At 520 A, this relay has to operate in $t_{6a} = 0.5 + 0.4 = 0.9$ s.

$PSM_{6a} = 520 \times (5/200) \times (1/2) = 6.5$ times. At 6.5 times and $t_{6a} = 0.9$ s, the time dial setting $= 0.37 \Rightarrow 0.40$.

Instantaneous setting $= 1.25(1\,025.15)(5/200) = 32.04 \Rightarrow 32$ A.

$$I_{\text{prim. trip}} = 32\,\text{A} \times \frac{200}{5} = 1\,280\,\text{A}$$

Relay 8:
Relay backs up relays 6 and 7 and should be coordinated with the slower of these two relays. Relay 7 has an instantaneous primary current setting of 1 100 A, equivalent to 27.5 A secondary current that is less than the setting of relay 6, and so the operating time of both relays is determined by this value.

For relay 7, $PSM = 1\,100 \times 5/200 \times 1/4 = 6.87$ times. At 6.87 times and with a time dial setting of 0.3, $t_{op} = 0.69$ s.

For relay 6, $PSM = 1\,100 \times 5/200 \times 1/2 = 13.75$ times. At 13.75 times and with a time dial setting of 0.4, $t_{op} = 0.42$ s.

Therefore, the operating time to give correct discrimination with relay 7 is $t_{8a} = 0.69 + 0.4 = 1.09$ s.

For back-up relay 8, the contributions to relay 6 from substations G and M are not considered. Only the infeed from the transformer has to be taken into account, so that $PSM_{8a} = 1\,100 \times (2\,170.34/3\,060.40) \times (5/300) \times (1/7) = 1.86$ times. At 1.86 times and $t_{8a} = 1.09$ s, the time dial setting $= 0.07 \Rightarrow 0.1$.

Here also, no instantaneous setting is applied to relay 8 for the reasons given in section 5.2.1. The maximum short-circuit current to be used for this relay is the current that flows from the 115-kV busbar to the 34.5-kV busbar for a fault on the latter, and $PSM_{8b} = 2\,170.34 \times (5/300) \times (1/7) = 5.17$ times. At 5.17 times and with a time dial setting of 0.1, $t_{8b} = 0.32$ s.

Relay 9:
This relay backs up relay 8 in a time of $t_{9a} = 0.4 + 0.32 = 0.72$ s.

$PSM_{9a} = 2\,170.34 \times (34.5/115) \times (5/250) \times (1/3) = 4.34$ times. At 4.34 times and $t_{9a} = 0.72$ s, the time dial setting $= 0.18 \Rightarrow 0.20$.

The instantaneous setting $= 1.25 \times 2\,170.39 \times (34.5/115) \times (5/250) = 16.28$ A $\Rightarrow$ 17 A.

$$I_{\text{prim. trip}} = 17 \times \left(\frac{250}{5}\right) = 850\,\text{A, referred to}115\,\text{kV}$$

The coordination curves of the relays associated with this system are shown in Figure 5.22 and summarised in Table 5.6.

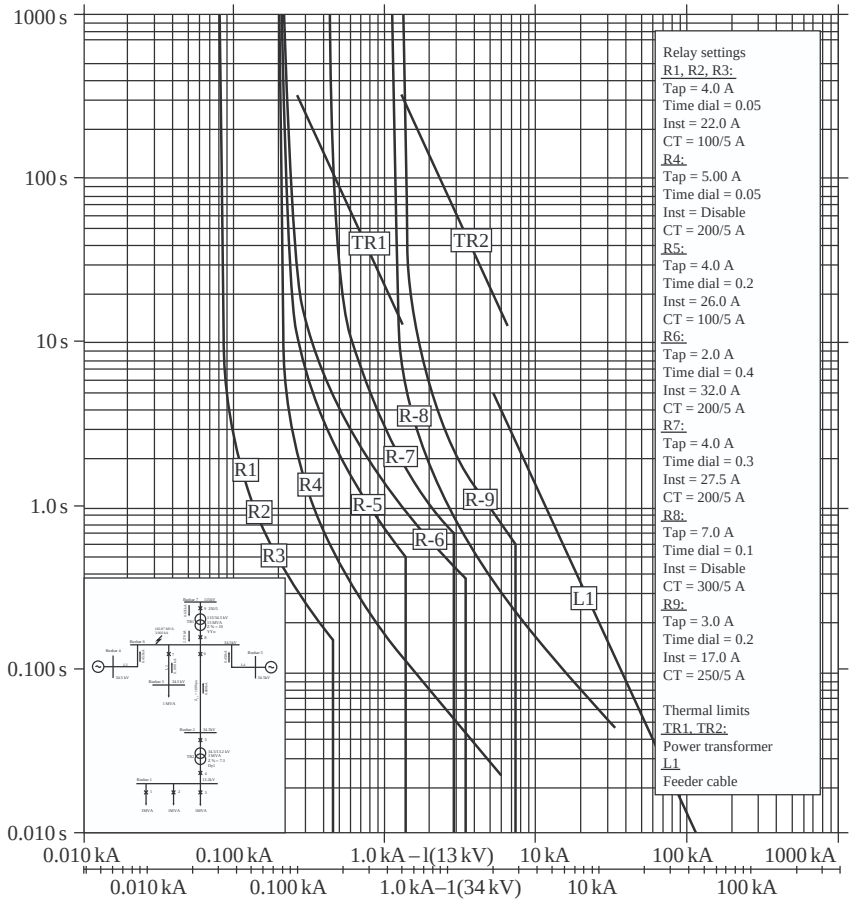


Figure 5.22 Relay coordination curves for Example 5.3

Percentage of 34.5-kV line protected by the instantaneous element of the overcurrent relay associated with breaker 6:

Given that

$$\% = \frac{K_s(1 - K_i) - 1}{K_i}$$

Table 5.6 Summary of settings for Example 5.3

Relay number	CT ratio	Pick-up (A)	Time dial	Instantaneous I_{sec} (A)
1–3	100/5	4	½	20.0
4	200/5	5	½	–
5	100/5	4	3	26.0
6	200/5	2	6	32.0
7	200/5	4	5	27.5
8	300/5	7	1	–
9	250/5	3	2	17.0

where

$$K_i = \frac{I_{sc.pick\text{-}up}}{I_{sc.end}} = \frac{1\,280}{1\,025.15} = 1.25$$

and

$$K_s = \frac{Z_{source}}{Z_{element}}$$

From the computer listing:

$$Z_f = \frac{V^2}{P} = \frac{(34.5)^2}{183.11} = 6.50$$

and

$$K_s = \frac{6.50}{12.93} = 0.50$$

Therefore,

$$\% = \frac{0.50(1 - 1.25) + 1}{1.25} = 0.70$$

so that the instantaneous element covers 70% of the line.

5.6 Coordination with fuses

When coordinating overcurrent relays, it may be necessary to consider the time/current characteristics of fuses that are used to protect MV/LV substation transformers. When a fuse operates, the circuit is left in an open-circuit condition until the fuse is replaced. It is therefore necessary to consider the case of preventing fuse operation because of the problems of replacing it after it operates, which is called fuse saving. In these cases, it may be preferable to forgo the selectivity of the protection system by not taking account of the fuse characteristic curve, so that the fuse will then act as a back-up.